

BeeAware: Application Usage Monitoring and Time Management System for Habit Improvement

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Abstract—Modern digital workspaces expose users to persistent algorithmic distractions, causing significant task-switching costs and cognitive fatigue. This paper presents BeeAware, a privacy-preserving, local edge-AI tracking ecosystem that transforms passive operating system window metadata into real-time productivity interventions. Adhering to the CRISP-DM data science lifecycle, the system captures active window text streams and classifies them into the four quadrants of the Eisenhower Matrix using a Term Frequency-Inverse Document Frequency (TF-IDF) feature engineering pipeline coupled with a localized Linear Support Vector Classifier (Linear SVC). To protect data privacy, all computations run locally, augmented by a lightweight MediaPipe face-detection tracking toggle. The machine learning architecture achieves a 98% macro accuracy and 0.98 F_1 -score on a custom, balanced corpus of 12,387 domain-specific activity entries. Usability tests among academic and technical professionals in the Mindanao region yielded a mean System Usability Scale (SUS) score of 69.29 and an ISO/IEC 25010 privacy score of 4.57 out of 5.00, demonstrating that client-side behavioral modeling can mitigate digital procrastination without compromising user privacy.

Index Terms—Data Science, CRISP-DM, Short-Text Classification, Linear SVC, Edge AI, Edge Privacy, Human-Computer Interaction, Behavioral Nudging.

I. INTRODUCTION

The global digital landscape has transformed personal computers into double-edged instruments of unprecedented economic productivity and relentless cognitive fragmentation. Globally, digital consumption has surged dramatically, with users averaging nearly seven hours of daily internet exposure spent navigating algorithmically optimized digital ecosystems [6], [11]. This continuous connectivity has given rise to a global attention economy that engineered platforms to systematically exploit human cognitive vulnerabilities, resulting in a documented decline in individual attention spans and psychological well-being [1], [10]. Modern workspaces frequently demand prolonged interaction with digital screens, yet these environments expose users to perpetual interruptions, inducing a state of chronic task-switching and cognitive fatigue [19]. This tension underscores a critical problem in Human-Computer Interaction (HCI): while digital connectivity is mandatory for modern workflows, it simultaneously degrades the focused concentration required to execute high-value cognitive tasks [12].

In the national context of the Philippines, these digital dependencies severely stress emerging remote working and

hybrid academic paradigms. Recent regional metrics show that the typical Filipino internet user maintains some of the highest daily media consumption rates globally, a trend that directly mirrors a sharp inflation in smartphone and internet dependency among students and the white-collar workforce [11]. While this hyper-connectivity supports the country's booming digital labor market, it introduces a severe systemic vulnerability: high-frequency digital distractions are closely linked to escalating rates of academic procrastination, psychological distress, and a notable drop in overall academic performance [8], [9], [20]. This digital disruption limits professional efficacy and destabilizes work-life balance within domestic environments, threatening organizational commitment across local public and private sectors [13]. Consequently, there is an urgent national need for scalable, localized tools that can protect productivity without compromising user privacy.

At the immediate local level, this crisis is visibly apparent within regional educational and administrative infrastructures. Regional directives, such as the Department of Education (DepEd) Davao Region Strategic Plan, have explicitly prioritized digital literacy alongside the cultivation of responsible digital citizenship to mitigate the erosion of student study habits caused by mobile phone and desktop dependencies [4], [5]. However, existing digital time-management interventions fall short of these objectives. Commercial tracking tools like RescueTime or Apple Screen Time operate on passive, historical charting mechanics that provide retroactive awareness rather than real-time behavioral guidance [33]. Furthermore, these systems require users to manually input and categorize their activities—a tedious administrative hurdle that often leads to rapid user abandonment [7]. More critically, modern users are increasingly hesitant to adopt tracking utilities that exfiltrate sensitive, second-by-second desktop logs to external cloud servers, creating a fundamental tension between productivity logging and individual data privacy.

To bridge these gaps, this paper presents **BeeAware**, a localized, privacy-preserving edge-AI tracking and behavioral coaching ecosystem. Built on the six phases of the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework, BeeAware transforms passive operating system telemetry into active behavioral interventions. The core architecture automatically intercepts active window title text streams and classifies them across the four quadrants of

the Eisenhower Matrix—Urgent/Important (Deep Work), Urgent/Not Important (Reactive Noise), Not Urgent/Important (Strategic Planning), and Not Urgent/Not Important (Distraction)—mitigating the cognitive biases that lead individuals to prioritize short-term trivialities over long-term goals [3], [25]. By executing text feature extraction via Term Frequency-Inverse Document Frequency (TF-IDF) alongside a localized, lightweight margin classifier, BeeAware delivers real-time, context-aware push notifications to gently nudge users back toward their intended workflows [14], [37]. Crucially, the entire machine learning inference loop executes locally on the host machine, ensuring absolute data privacy.

The principal contributions of this research are structured as follows:

- **Privacy-Preserving Edge Architecture:** We design and implement a zero-cloud, client-side tracking system that processes low-level operating system telemetry entirely within local runtime boundaries, eliminating external data vulnerability risks.
- **Context-Aware Multi-Class Classification:** We develop a lightweight Natural Language Processing (NLP) pipeline optimized for high-dimensional, short-text window titles to dynamically map active user states into behavioral matrix coordinates.
- **Real-Time Persuasive Intervention Engine:** We integrate a localized feedback mechanism that translates real-time classification changes into non-intrusive behavioral nudges, encouraging sustainable habit formation.
- **Empirical Usability Evaluation:** We validate the system’s operational viability and user acceptance through structured System Usability Scale (SUS) assessments within a live environment [31], [34].

II. RELATED WORK

A. The Digital Attention Crisis and Global Productivity Paradox

The modern digital landscape is increasingly defined by the attention it garners, where software applications are systematically engineered to maximize user engagement through psychological means [1], [10]. While digital technologies are initially deployed as catalysts for organizational and personal efficiency, their invasive nature often triggers a phenomenon known as the “productivity paradox” [33]. Empirical observations reveal that the average user checks their personal device over 96 times a day, frequently falling into a digital “rabbit hole” that consumes hours of non-essential time on superficial media platforms [12]. A primary research gap identified on a global scale is the lack of active intervention systems within consumer software design; although users maintain an abstract awareness of their soaring screen-time metrics, they lack the real-time, structured feedback mechanisms required to disrupt deeply ingrained, destructive behavioral habits [14], [37].

B. National and Regional Context: Digital Literacy and Fatigue

This attention crisis manifests with heightened severity within the Philippines, which continually registers as a global leader in digital media consumption. Empirical reporting indicates that the average daily internet exposure among the domestic population exceeds 10 hours [11]. This high level of connectivity frequently correlates with adverse behavioral and psychological outcomes. Academic dependencies on mobile devices and personal computers significantly contribute to distracted study habits, reduced cognitive focus, and prolonged cycles of academic and professional procrastination [5], [9], [20].

On a regional scale within Mindanao, studies at institutions such as Mindanao State University (MSU) Marawi have further documented a significant correlation between social media over-dependency and severe academic performance drops among students [8]. Furthermore, within expanding urban centers like Davao City and the Northern Mindanao region, the sudden transition to digital-first workflows has introduced widespread digital fatigue [13]. In these environments, the system-level inability to distinguish between high-priority workplace operations and low-priority social notifications leads to decreased job satisfaction and emotional burnout [13]. While regional educational bodies have initiated strategic directives, such as the Department of Education (DepEd) Davao Region Strategic Plan, these programs remain pedagogical and instructional rather than technological [4]. This leaves a critical vacancy for automated systems capable of actively managing and mitigating these digital habits at the hardware interface level.

C. Limitations of Existing Digital Well-Being Tools

Current market leaders, such as Apple Screen Time and Android Digital Wellbeing, provide what is categorized in human-computer interaction (HCI) literature as “passive reporting” [33]. These utility frameworks focus strictly on aggregating historical usage data and executing binary application blocking. However, behavioral studies suggest that these rigid, restrictive-only models often induce heightened user frustration or trigger a “rebound usage” effect, wherein users over-consume distracting media or abandon the tool entirely once the artificial time block expires [7], [10]. Furthermore, comparative analyses show that existing tools lack contextual intelligence. They quantify productivity using broad, hard-coded application categories but fail to recognize when a user is spending excessive time on low-priority tasks within an otherwise productive environment (such as a browser tab running non-essential streams), highlighting the necessity for a nuanced, importance-based classification framework.

D. The Eisenhower Matrix as a Classification Framework

To establish an objective multi-class target space, researchers leverage the Eisenhower Matrix—a classic task-management tool that categorizes activities into a four-quadrant grid predicated on the distinct thresholds of ur-

gency (tasks demanding immediate reaction) and importance (tasks aligning with long-term missions) [3]. In contemporary tracking applications, these quadrants are mapped to distinct behavioral states: Q1 (Urgent & Important; Crisis/Do), Q2 (Not Urgent & Important; Growth/Plan), Q3 (Urgent & Not Important; Noise/Interruptions), and Q4 (Neither Urgent nor Important; Play/Distractions). The effectiveness of this matrix lies in its mathematical capacity to mitigate the “mere-urgency effect,” a cognitive bias where individuals systematically prioritize short-term urgent tasks over high-value important ones, regardless of their objective long-term outcomes [25]. By combining this psychological model with algorithmic classification, workstations can automate honest task evaluation, combat decision fatigue, and actively guide users toward highly productive Q2 behaviors.

E. Edge Computing and Webcam-Based Workstation Monitoring

The use of webcam-based presence detection for workstation monitoring has gained significant traction, driven by the need for real-time occupancy data without relying on intrusive or high-cost external hardware arrays. MediaPipe provides a robust, cross-platform framework for building perception pipelines capable of running single-shot face detection models at sub-millisecond latency directly on a local CPU or GPU [21], [35]. These facial landmark detection approaches remain highly robust under background deployment conditions, maintaining an accuracy profile exceeding 97% across diverse ambient lighting conditions with minimal computational overhead on consumer-grade machines [35], [36]. Crucially, local edge execution serves as the privacy-preserving gold standard in ambient monitoring [23]; because all frames are processed locally and pixel data is immediately discarded post-inference, raw image files never leave the local machine, matching the strict privacy constraints of the *BeeAware* system architecture.

F. TF-IDF Feature Space and Linear Support Vector Classifiers

Extracting contextual intent from unstructured, high-dimensional short text—such as operating system window titles or browser tab strings—remains a fundamental challenge in context-aware computing [17]. To overcome this, text processing pipelines employ Term Frequency-Inverse Document Frequency (TF-IDF) combined with n -gram vectorization as a statistical filtering engine [18]. TF-IDF assigns heavy statistical weights to high-signal, context-heavy keywords (e.g., `PostgreSQL`, `GitLab`) while suppressing low-signal, generic terms, enabling systems to accurately isolate a user’s semantic intent [18].

When text is converted into these sparse, high-dimensional vector spaces, traditional models like Decision Trees often struggle with overfitting or intensive compute overhead. Conversely, a Linear Support Vector Classifier (Linear SVC) is highly optimal for text classification due to its innate mathematical ability to seamlessly manage sparse features while maintaining robust generalization boundaries [29]. Because

short text headers are frequently linearly separable in high-dimensional domains, a linear kernel offers rapid training and sub-millisecond inference capabilities [29]. This allows a multi-class Linear SVC to establish predictable, immediate decision boundaries between specific digital tasks and their corresponding Eisenhower quadrant assignments without exhausting host workstation resources [30].

G. The Role of Custom-Labeled Corpora in Domain-Specific NLP

A fundamental barrier in natural language processing is the semantic gap between generic vocabulary and domain-specific intent. Off-the-shelf text datasets consistently underperform when tracking specialized workflows, causing classifiers to misinterpret technical system titles (e.g., `VSCode`, `localhost`) as generic noise or misclassifying their urgency thresholds [26], [28]. Consequently, the assembly of a localized, custom-labeled corpus is a recognized prerequisite for training supervised classifiers to recognize unique workflow signatures [26]. A manually curated corpus serves as the empirical ground truth that guides the TF-IDF vectorizer to position optimal decision boundaries [27]. By explicitly pairing local technical strings with targeted psychological indicators (e.g., pairing `System Error` with Q1 and `Social Media` with Q4), the resulting domain-specific corpus provides the multidimensional context required for an AI model to operate efficiently as a real-time behavioral coach [26].

H. Interactive Dashboards and the Intention-Behavior Gap

Accurately classifying user activity is only half of the required intervention; the resulting telemetry must be presented visually to overcome cognitive friction and trigger behavioral realization. In human-computer interaction (HCI) research, studies show that passive, static data reporting fails to bridge the “intention-behavior gap,” frequently overwhelming users and leading to intervention blindness [15], [32]. This limitation necessitates a shift toward dynamic, interactive visual analytics. Real-time dashboards that continuously render graphical analytics significantly increase personal awareness and allow for immediate, targeted self-correction the moment a user slips into a state of distraction [16], [22]. By utilizing localized, interactive visualization libraries like Plotly to project active time tracking metrics directly onto a live Eisenhower Matrix representation, a user can synthesize raw digital inputs into an engaging, structured roadmap for habit modification [33].

I. Research Gap and Project Synthesis

Despite significant advancements in User Activity Recognition (UAR) and text categorization, a critical cross-disciplinary research gap remains at the intersection of automated short-text processing, edge computational privacy, and active human-computer interaction (HCI) coaching frameworks. Current software solutions suffer from a strict architectural and functional split:

- **The Passive vs. Active Coaching Gap:** Commercial digital well-being platforms rely almost entirely on “passive reporting” frameworks that summarize historical

usage data through retroactive graphs [33]. When active control is attempted, these systems deploy rigid, binary application blocks that induce user frustration and trigger rebound media over-consumption once restrictions expire [7], [10]. There is a definitive lack of intermediate, context-aware systems that utilize real-time persuasive “nudges” to modify behavior without enforcing disruptive, hardcoded application lockouts [14], [37].

- **The Semantic and Contextual Blindness Gap:** Traditional time trackers analyze raw application process titles or generic domain names uniformly, completely missing the underlying cognitive intent of the user [17]. For example, existing architectures uniformly categorize a web browser domain as either wholly productive or completely distracting. They fail to parse high-dimensional, short textual metadata—such as evolving desktop window titles or browser tab headers—leaving them unable to detect when a user is drifting into low-priority tasks within an otherwise productive application environment [18].
- **The Edge Privacy vs. Computational Overhead Dichotomy:** Advanced natural language processing models capable of parsing short texts often rely heavily on cloud-connected infrastructures or massive language networks [26]. This introduces severe data privacy liabilities, as users are understandably hesitant to exfiltrate highly sensitive, second-by-second desktop window strings to remote servers. Conversely, current local-only tracking mechanisms remain highly simplistic and rule-based because running intensive, non-linear machine learning classifiers directly on a monitored workstation can exhaust local hardware resources and compromise system performance [29], [30].
- **Geographical and Operational Specificity Vacuum:** Existing digital well-being research is primarily tailored to generic Western or highly globalized corporate workflows, creating an empirical vacancy for localized, target-specific interventions [28]. Specifically, within regions experiencing rapid structural shifts toward hybrid education and remote freelancing—such as the Mindanao region in the Philippines—users face a distinct combination of digital fatigue, academic procrastination, and institutional demands for digital literacy [4], [9], [13]. Current literature lacks an empirical framework that actively evaluates how automated edge classifiers perform when subjected to localized, domain-specific student and professional workflow vocabularies [26], [27].

BeeAware directly bridges these interrelated research gaps. Moving beyond the historical charting limits of standard utilities, this system introduces an active, local-only edge architecture that pairs text-vectorization with machine learning to provide immediate cognitive support. To resolve the tension between processing depth and local hardware strain, this study implements a highly optimized, high-dimensional TF-IDF feature vector space evaluated by a regularized Linear Support

Vector Classifier (Linear SVC)—benchmarked directly against an ensemble Random Forest framework to identify the mathematical sweet spot for local execution [27], [29].

Furthermore, by mapping these real-time streams onto the psychological decision boundaries of the Eisenhower Matrix, the system converts raw, high-frequency text telemetry into rich visual telemetry—such as task “switching intensity” and quadrant distributions rendered through an interactive Plotly dashboard [3], [16], [25]. Ultimately, by testing this unified framework against real-world scenario sessions validated by the System Usability Scale (SUS), this research provides a scalable, privacy-preserving blueprint for technological habit management explicitly optimized for the modern digital workflows of students and professionals within the region [31], [34].

III. METHODOLOGY

The development and evaluation of *BeeAware* follow the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework, translating raw human-computer interaction (HCI) telemetry into a multi-class machine learning classification and real-time behavioral coaching pipeline [33]. The system architecture operates strictly on the client-side edge to guarantee user data privacy, processing high-frequency interface metadata entirely within local workstation runtime boundaries [23].

A. Business and Operational Understanding

The primary objective of this phase is to map qualitative productivity challenges—such as academic procrastination and digital fatigue in the Mindanao region—into an objective, automated classification task [8]. Rather than treating screen-time as a monolithic, daily aggregate metrics, the system models user focus dynamically using the four quadrants of the Eisenhower Matrix [3]. Tasks are partitioned across a dual-axis threshold of urgency and importance to separate high-value strategic execution from low-value distraction loops:

- **Quadrant 0 (Q0 - Urgent):** Immediate, high-consequence crisis engineering and system-level operations demanding instantaneous focus [3].
- **Quadrant 1 (Q1 - Deep Work):** Non-urgent but highly important core tasks matching long-term development, academic research, and deep composition [3].
- **Quadrant 2 (Q2 - Reactive Noise):** Urgent but non-important systemic interruptions, including communication pop-ups, administrative notifications, and transactional emails [3].
- **Quadrant 3 (Q3 - Distraction):** Low-value, non-urgent, and non-important media binging, social scrolling, and interactive gaming that compromise long-term productivity [3].

B. Data Understanding and Custom Activity Corpus Creation

To overcome the limitations of off-the-shelf text datasets that display a critical semantic gap when processing specialized professional and local academic workflows, a domain-specific custom-labeled corpus was engineered [26].

The dataset compilation followed a multi-source extraction pipeline targeting raw operating system interface strings [28]. High-frequency software engineering tokens (e.g., `VSCode`, `PostgreSQL_Lab`, `8086_Assembly`) alongside local organizational metadata were programmatically captured under naturalistic workstation environments [26]. Multimedia elements were gathered via targeted browser categorical schemas (`URL_classification.csv`) to capture educational video logs and separate them from purely recreational streams [18].

Following data deduplication and cross-dataset parsing, conflicting entries were resolved to establish a clean empirical ground truth. The finalized consolidated gold-standard corpus, designated as `V3beeware_master_training.csv`, comprises a total of 12,387 unique window title entries symmetrically balanced across the four targeted behavioral classes: Q0 (3,005 entries), Q1 (3,414 entries), Q2 (3,121 entries), and Q3 (2,847 entries).

C. Data Preparation and Contextual Feature Engineering

Raw user desktop tracking strings are notoriously sparse, fragmented, and noisy. Prior to statistical tokenization, each captured record is passed through a unified feature construction function that expands the raw window string with a 5-dimensional categorical context array: process name (`process_name` using `psutil` tracking), tracking origin (`source_type`), application class (`app_family`), browser domain extraction (`browser_domain`), and sub-topic content categorization (`browser_category`).

These fields are structurally concatenated using a pipe delimiter to form a dense composite semantic text string known as the *feature text*, written mathematically as:

$$\text{FeatureText} = P_{\text{name}} \parallel S_{\text{type}} \parallel A_{\text{fam}} \parallel B_{\text{dom}} \parallel B_{\text{cat}} \parallel W_{\text{title}} \quad (1)$$

A concrete instantiation of this preparation sequence transforms a raw browser window into the following unified string: `chrome | browser | browser | Education | khanacademy.org | Khanacademy.org - Browser`. This composite feature structure ensures that the downstream vectorization pipeline interprets the localized words within the explicit runtime context of the operating system application environment.

D. Feature Extraction via TF-IDF Vectorization

The unstructured, multi-dimensional feature text composite strings are translated into sparse numerical arrays utilizing a Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer with integrated word-level n -gram constraints [18]. The mathematical mapping assigns local statistical weights to terms, maximizing the signal of distinctive workspace keywords while penalizing highly frequent, low-signal tokens across the corpus [18]. The weight for a term t within a document d inside a corpus of size N is computed using the standard formulation:

$$\text{TF-IDF}(t, d, D) = \text{tf}(t, d) \times \log \left(\frac{1 + N}{1 + \text{df}(t)} \right) + 1 \quad (2)$$

where $\text{tf}(t, d)$ represents the raw token frequency in document d , and $\text{df}(t)$ signifies the document frequency of term t across the entire collection. This creates a high-dimensional feature hyperspace where linearly separable boundaries can be mapped with micro-second tracking fidelity [29].

E. Modeling Framework and Classifier Benchmarking

To meet the strict client-side constraints of zero-cloud processing without introducing compute-induced desktop latency that disrupts user workflow, a highly optimized Linear Support Vector Classifier (Linear SVC) was implemented [29], [30]. The Linear SVC model explicitly fits a multi-class configuration to determine optimal separating hyperplanes within the high-dimensional TF-IDF vector space, maximizing the margin boundary between the different Eisenhower quadrants [29]. The optimization function minimizes squared hinge loss with L2 regularization:

$$\min_{w, b} \frac{1}{2} w^T w + C \sum_{i=1}^n \max(0, 1 - y_i(w^T \phi(x_i) + b))^2 \quad (3)$$

To empirically validate the structural choice of a margin-based linear separator over deep ensemble frameworks for workstation monitoring, the Linear SVC model is comprehensively benchmarked head-to-head against an ensemble Random Forest Classifier (comprising 100 independent decision estimators). Performance profiles are evaluated rigorously across standard criteria: overall macro accuracy, quadrant-specific precision, tracking recall, and multi-class F1-scores on an independent, stratified 20% held-out evaluation subset (2,370 testing validation strings).

F. System Integration, Edge-Vision Layer, and Plotly Interface

The trained machine learning models are serialized via `joblib` and integrated into a background client application tracking daemon written in Python. The input tracking loop uses low-level native hooks via `win32gui` and `psutil` to sample active desktop metadata at a constant frequency of 1 Hz. To ensure tracking integrity and filter out false-positive idle states (such as a user walking away from an open workstation), a parallel edge-vision layer is deployed [?]. This layer utilizes the MediaPipe Perception pipeline to run a lightweight, single-shot face detection model directly on the local frame buffer of the integrated webcam [21], [35].

Processing operates strictly on local volatile memory; pixel data is evaluated frame-by-frame at sub-millisecond speeds and immediately discarded post-inference, returning a binary occupancy state signal (present = 1, absent = 0) to the tracking core. If the signal registers an absence state, the application tracking loop pauses instantly, preventing data corruption from unmonitored desktop sessions. The real-time telemetry streams are logged continuously into a local, persistent cryptographic CSV activity ledger [33]. For the active user interface layer, these logs are parsed using an interactive data visualization dashboard engineered via the Plotly framework [33]. The diagnostic engine calculates and visualizes structural time metrics, quadrant allocation, application duration logs, and a

dynamic metric termed *switching intensity*—which mathematically tracks the frequency of rapid shifts between distinct task quadrants to quantify the onset of focus fragmentation and digital attention fatigue [16], [33].

G. Experimental Evaluation and Usability Framework

The human-computer interaction (HCI) viability and operational tracking fidelity of the integrated system were evaluated through an empirical, single-session scenario-based usability test under naturalistic working environments [31]. The experimental cohort comprised 14 active respondents recruited across major higher-education institutions within the Mindanao region, including Mapúa Malayan Colleges Mindanao, Ateneo de Davao University, the University of Southeastern Philippines (USEP), the University of Mindanao, and the University of the Immaculate Conception.

Participants completed structured, pre-defined workflow tracks designed to simulate common academic tasks, digital research, communication overhead, and distraction prompts while the background engine logged local telemetry. Following the tracking run, user experience dimensions were quantitatively captured using the standardized System Usability Scale (SUS) to evaluate software discoverability and system integration [31], [34]. System acceptance and psychological friction metrics were evaluated via the Technology Acceptance Model (TAM), alongside ISO/IEC 25010 standard dimensions focusing on security and local privacy trust architecture, and a custom NASA-TLX-derived cognitive assessment matrix evaluating the framework’s active impact on mitigating procrastination habits and digital fatigue indicators.

IV. RESULTS AND DISCUSSION

This section presents the empirical evaluation of the *BeeAware* system, focusing on the classification fidelity of the underlying machine learning models, computational efficiency on the edge, and human-computer interaction (HCI) metrics derived from the scenario-based usability evaluation.

A. Machine Learning Model Performance and Benchmarking

The primary objective of the modeling phase was to maximize classification accuracy while minimizing inference latency to ensure frictionless background execution on consumer workstations. The Linear Support Vector Classifier (Linear SVC) and the ensemble Random Forest Classifier were evaluated head-to-head on the 2,370 stratified validation samples extracted from the `V3beeware_master_training_v3.csv` gold-standard corpus.

The Linear SVC model demonstrated exceptional predictive capability, achieving an overall macro accuracy of **98%** and a macro F_1 -score of **0.98**. Per-class performance metrics are structured in Table I.

Analysis of the multi-class confusion matrix reveals perfect classification isolation (1.00 precision and recall) for Quadrant 0 (Urgent) and Quadrant 2 (Reactive Noise). This deterministic boundary is attributed to the presence of highly unique,

TABLE I
CLASSIFICATION PERFORMANCE METRICS FOR LINEAR SVC

Quadrant	Precision	Recall	F_1 -Score	Support
Q0 (Urgent)	1.00	1.00	1.00	601
Q1 (Deep Work)	0.96	0.97	0.96	683
Q2 (Reactive)	1.00	1.00	1.00	624
Q3 (Distraction)	0.96	0.95	0.95	562
Macro Avg	0.98	0.98	0.98	2,370

process-specific metadata tokens extracted via `win32gui` and `psutil` (e.g., terminal executions, local administrative network ports, and enterprise messaging headers) which exhibit zero semantic overlap with recreational software signatures.

Minor token confusion occurred strictly between Quadrant 1 (Deep Work) and Quadrant 3 (Distraction), where the model misclassified 24 instances. This overlap is due to ambiguous web browsing window titles where specific informational platforms (e.g., certain video-sharing sub-domains or community forums) contain identical linguistic tokens but serve divergent contextual roles depending on the user’s instantaneous task orientation. Despite this minor variance, the margin-based hyperplane boundaries proved highly robust, outperforming the ensemble Random Forest benchmark, which suffered from overfitting on short, high-frequency window strings and exhibited an increased local inferential footprint.

B. Computational Efficiency and Edge Privacy Integrity

A core constraint of the *BeeAware* framework is local execution. Edge-vision validation via the MediaPipe perception layer maintained stable tracking throughput, yielding a binary presence state matrix within sub-millisecond ranges [21], [35]. Local CPU utilization remained consistently below 2.4% during active sampling intervals, and volatile memory allocations for the serialized Linear SVC model and active dataframe logs did not exceed 42 MB. This confirms that embedding a continuous linear text classification loop alongside an edge computer-vision toggle achieves lightweight monitoring without inducing application latency or frame drops during demanding tasks like compilation or software engineering [23].

C. System Usability and Technology Acceptance

The system’s operational viability was contextualized through user evaluation among 14 active academic and technical professionals across higher education institutions in the Mindanao region.

TABLE II
DISTRIBUTION OF EVALUATION SCORES ACROSS USABILITY DIMENSIONS

Evaluation Metric Domain	Mean Empirical Score
System Usability Scale (SUS)	69.29 (Categorized as “Good”)
TAM: Perceived AI Accuracy	3.86 / 5.00
ISO/IEC 25010: Privacy/Security	4.57 / 5.00

The evaluation yielded a mean System Usability Scale (SUS) score of **69.29**, positioning *BeeAware* within the “Good” usability classification threshold [34]. This indicates that while the system integrates cleanly into local startup environments, additional engineering is required to polish user configurations and dashboard discoverability.

Under the Technology Acceptance Model (TAM) dimensions, the system scored a high mean rating of **3.86 out of 5.00** for perceived AI predictive accuracy, matching the objective numerical validity observed during text verification. Crucially, the system scored its highest rating under the ISO/IEC 25010 security and structural privacy sub-scale (**4.57 out of 5.00**). Users noted that the absolute absence of cloud-based web endpoints or external database mirrors fostered deep trust, effectively mitigating the surveillance anxiety typically associated with automated productivity-tracking tools.

Feedback from the NASA-TLX-derived cognitive assessment indicated that the real-time Plotly interactive dashboard—specifically the *switching intensity* index—successfully made cognitive fatigue visible to the users [16]. This visibility prompted participants to initiate intentional breaks when entering high-frequency task-switching loops, demonstrating the system’s viability as an active tool against digital procrastination.

V. CONCLUSION AND RECOMMENDATIONS

This study successfully designed, implemented, and validated *BeeAware*, a localized, privacy-first productivity optimization system utilizing the CRISP-DM framework [33]. By extracting native interface metadata via `win32gui` and process telemetry via `psutil`, and combining it with a lightweight MediaPipe edge-vision occupancy layer, the system establishes a secure, client-side tracking architecture [21], [35].

The engineered Linear SVC classification model achieved an overall accuracy of **98%** and a macro F_1 -score of **0.98** when evaluated on a custom domain-specific corpus of 12,387 highly contextual tokens. This model effectively maps ambiguous user actions into the four quadrants of the Eisenhower Matrix in real time. Empirical user testing across key institutions in the Mindanao region confirmed high structural privacy trust (**4.57/5.00**) and solid system usability (**69.29 SUS**), validating the approach of deploying lightweight machine learning models at the edge to combat digital fatigue and procrastination without compromising user privacy [31], [34].

To advance the development of local behavioral computing systems, the following future research tracks are recommended:

- **Dynamic Adaptive Re-vectorization:** Future iterations should integrate an incremental, localized on-the-fly learning loop allowing users to manually re-assign misclassified window tokens directly from the Plotly dashboard interface, automatically retraining the local hyperplane boundaries without requiring a full manual compilation of the training corpus.
- **Cross-Platform Core Telemetry:** Expanding the underlying operating system hook framework beyond the native Windows `win32gui` subsystem to native Linux (X11/Wayland) and macOS (Quartz API) wrapper layers to offer uniform productivity monitoring across diverse software engineering ecosystems.
- **Temporal Trend Analytics:** Transitioning the analytical engine from isolated 1 Hz data frames into sequential modeling architectures (e.g., hidden Markov structures or temporal window attention blocks) to evaluate semantic changes over extended periods, providing deep insights into behavioral evolution and long-term habits.

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